JEREMY MATTATHIAS MBOE

AI Engineering Student

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SUMMARY

Al Engineering student focused on the geometry and mathematics of generative models, including score-based diffusion, flow matching, and representation learning, with applications to scientific and medical problems. Experienced in implementing non-standard training methods in PyTorch from scratch, including surrogate-gradient spiking neural networks and physics-informed neural networks for ill-posed inverse problems. Currently extending this work toward large language models and foundation models for scientific applications.

PROJECTS & RESEARCH

SwaraTTS: Latent Flow Matching (Avalon AI X RISTEK UI) 02/2026 – Present
Deep Learning Researcher (Diffusion Models & Flow Matching Division)
- Collaborated within a multidisciplinary team of engineers and researchers to develop a low-latency neural speech synthesis system.
- Researched and implemented Score-Based Diffusion Models and Latent Flow Matching (LFM) to optimize the generative trajectory of neural speech.
- Explored the trade-offs between Continuous Normalizing Flows (CNF) and diffusion-based paradigms to achieve state-of-the-art sampling efficiency.

Foundational Neuromorphic Vision Research 04/2026 – Present
Avalon AI Research Assistant
- Engineered a custom SNN training pipeline in PyTorch entirely from scratch custom surrogate gradient functions, temporal encoders, and spike-based layers, with no external SNN framework constructing a usable gradient path through a non-differentiable spiking nonlinearity.
- Researching Spiking Neural Networks with Izhikevich neuron dynamics and temporal encoding for image classification, achieving 76.6% accuracy on N-MNIST.

AVID-MM: Audio-Visual Identity Disentanglement 04/2026 – Present
Lead Deep Learning Researcher
- Developed a robust cross-modal speaker verification system designed to identify individuals in multilingual environments even when one modality (audio or visual) is missing or corrupted.
- Engineered a multimodal embedding pipeline using ECAPA-TDNN for voice features and pretrained face backbones, achieving a jump from 59% to 93%+ accuracy on the POLY_1 challenge setup.

Physics-Informed Lung Reconstruction & TB Cavity Segmentation 04/2026 – Present
Deep Learning Researcher | Multimodal TB Active Case Finding Project
- Developing a high-impact research pipeline for 3D CT Reconstruction from Triplanar X-rays (AP, PA, and Lateral) to automate Tuberculosis diagnostic screening.
- Researched Physics-Informed Neural Networks (PINNs) and Neural Attenuation Fields (NAF) to parameterize the Beer-Lambert law, solving the ill-posed inverse problem of sparse-view tomography.
- This project is part of The TB Detection and Diagnostic Big Project for Detection and Diagnostic.

REXTRA: LLM Inference Backend 03/2025 – 02/2026
Backend AI Engineer
- Built and maintained REST API infrastructure serving LLM inference, handling request routing, retries, and failover across multiple external model providers.
- Optimized request handling and connection management between backend services and provider APIs to reduce end-to-end inference latency.

Multimodal TB Cough Detection & Diagnostic System 08/2025 – Present
Machine Learning Developer
- Developed an AI-driven Tuberculosis detection system utilizing LSTM integrated with acoustic signal processing.
- Awarded 1st Place & Best Scientific Paper at the National Electrolympic UPNVJ and presented technical methodologies to a panel of academic experts.

EEG Signal Analysis and Binary Encoding 01/2026 – Present
Pusat Studi Kecerdasan Artifisial | AI Researcher
- Engineered an end-to-end signal processing pipeline to classify ALS vs. Normal subjects using 1,809 clinical EDF recordings.
- Developed a specialized feature extraction pipeline for segmented EEG tasks (Resting, Thinking, Typing), computing time domain features (MAV, Variance, STD) across specific subbands (Delta through Gamma).

EXPERIENCE

Laboratory Assistant 05/2026 – Present | Surabaya, Indonesia
Intelligent Computing and Vision Laboratory
- Collaborate with laboratory researchers, faculty members, and clinicians from Rumah Sakit Universitas Airlangga on medical AI research, contributing to the development and evaluation of deep learning-based medical image segmentation pipelines for kidney cyst segmentation.
- Develop and organize practicum modules and instructional materials for AI-related courses, including Machine Learning, Data Mining, and related computational intelligence topics, supporting hands-on student learning.

Research Assistant 02/2026 – Present | Surabaya, Indonesia
Diagnostic and Prognostic Laboratory (D&P Lab) | AI Division
- Developing a 3D CT Reconstruction pipeline from triplanar X-rays using Physics-Informed Neural Networks (PINNs) to solve underdetermined inverse problems in TB diagnostic imaging.
- Developed an automated TB Cough Diagnostic System utilizing LSTM architectures and acoustic feature extraction, achieving high-accuracy screening in clinical environments

Probability and Statistics Teaching Assistant - Department of Informatics Engineering 02/2026 – Present | Surabaya, Indonesia
Institut Teknologi Sepuluh Nopember
- Facilitated student learning in Probability Theory, Statistical Inference, and Stochastic Processes.

Software Developer Intern 01/2026 – 02/2026 | Surabaya, Indonesia
Pusat Studi Kecerdasan Artificial dan Teknologi Digital Institut Teknologi Sepuluh Nopember
- Executed EEG signal preprocessing and multi-dimensional feature extraction, focusing on specific channels, frequency bands, and sub-bands for neurological data analysis.
- Engineered the backend processing pipeline for a team of 4, enabling the seamless integration of signal processing scripts into a functional web application.
- Developed and evaluated Machine Learning classifiers to interpret EEG patterns for diagnostic support.

AWARDS

Top 50, Social Project Program Innovillage (2025) 04/2026
INNOVILLAGE Telkom Indonesia

1st Place & Best Scientific Paper National Electrolympic UPNVJ (2025). 06/12/2025
Universitas Pembangunan Nasional Veteran Jakarta
Recognized for the technical research and paper presentation of the TB Cough Detection System.

ORGANISATIONS

Avalon AI Group 11/2025 – Present | Surabaya, Indonesia
Computer Vision Specialist
- Intensive specialized training in Computer Vision, mastering state-of-the-art architectures and implementation strategies.

EDUCATION

Bachelor of Artificial Intelligence Engineering 08/2024 – Present | Surabaya, Indonesia
Institut Teknologi Sepuluh Nopember
- GPA: 3.86/4.00, Relevant Courses: Probability and Statistics, Computational Intelligence, Data Mining, Machine Learning, Deep Learning, Natural Language Processing, Digital Image Processing.

SKILLS

Programming Languages: Python, C++, JavaScript, and SQL (PostgreSQL).

AI & Machine Learning: PyTorch (from-scratch training loops, custom autograd and surrogate gradients), Score-Based Diffusion, Flow Matching & Continuous Normalizing Flows, Physics-Informed Neural Networks, Spiking Neural Networks, LLM Inference & Serving, Scikit-Learn, TensorFlow.

Mathematical Foundations: Linear Algebra, Probability & Stochastic Processes, Stochastic Differential Equations, Optimal Transport & Flow Matching, Variational Inference, and Inverse Problems.

Languages: English (Professional), Indonesian (Native).

CERTIFICATES

Dicoding : Machine Learning & Deep Learning Dev Certification for Machine Learning with TensorFlow

- Completed a hands-on program covering supervised and unsupervised learning, model training and evaluation, data preprocessing, and model deployment using TensorFlow.